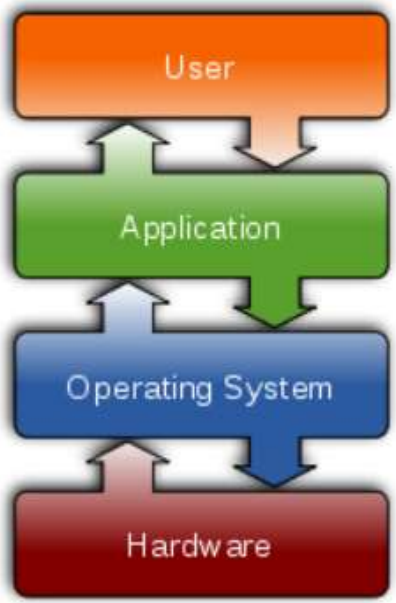
Lesson 3

# Introduction to Linux

# Operating systems

- **operating system**: Manages activities and resources of a computer.
  - software that acts as an interface between hardware and user
  - provides a layer of abstraction for application developers

- features provided by an operating system:
  - ability to execute programs (and multi-tasking)
  - memory management (and virtual memory)
  - file systems, disk and network access
  - an interface to communicate with hardware
  - a user interface (often graphical)




- **kernel**: The lowest-level core of an operating system.

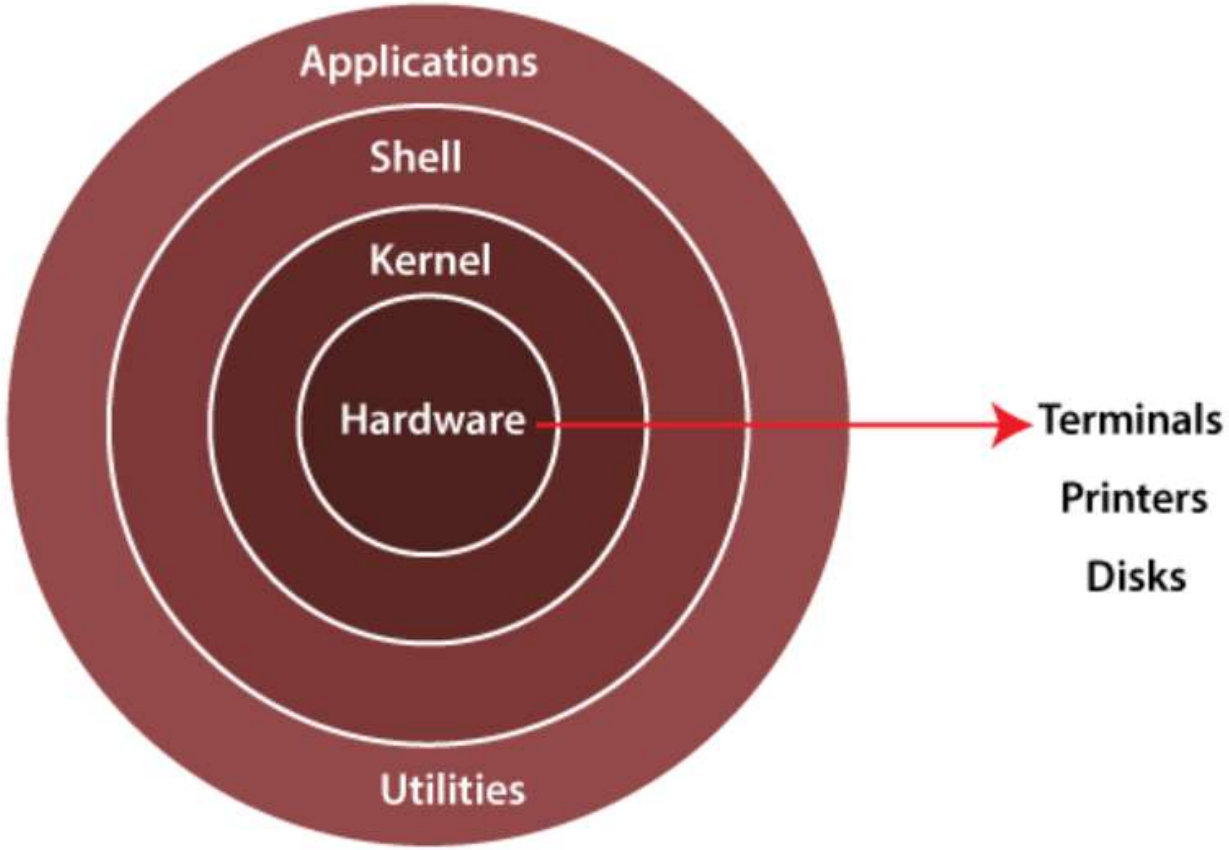
# UNIX/LINUX History

- The UNIX/LINUX operating system was born in the late 1960s. It originally began as a one man project led by Ken Thompson of Bell Labs, and has since grown to become the most widely used operating system.
- In the time since UNIX/LINUX was first developed, it has gone through <u>many</u> different generations and even mutations.
  - Some differ substantially from the original version, like Berkeley Software Distribution (BSD) or Linux.
  - Others, still contain major portions that are based on the original source code.
- An interesting and rather up-to-date timeline of these variations of UNIX can be found at http://www.levenez.com/unix/history.html.

# Characteristics/Features of LINUX

- **Multi-user** – Multiple users can access system resources at the same time.
- **Multi-tasking** – Multiple jobs can run at the same time.
- **Multiprogramming –** Multiple applications can run at the same time.
- **Large Number of Applications** – there are an enormous amount of applications available for UNIX operating systems.
- **Open Source Operating System** - of all of the applications available under UNIX, many of them are free. The compilers and interpreters that we use in most of the programming courses here at UMBC can be downloaded free of charge. Most of the development that we do in programming courses is done under the Linux OS.
- **Portable** – All programs can run on different system in same functionality.
- **Shell –** Provides interface to the users to run their programs.
- **Internet Development** - Much of the backbone of the Internet is run by UNIX servers. Many of the more general web servers run UNIX with the Apache web server - another free application.

# Linux Architecture



# Linux Architecture Cont’d…

1. **Hardware**: It is a combination of all peripherals associated with the system. For example: RAM, CPU and Hard disk etc..
2. **Kernel**: It is the core part of the operating system and manages the CPU, memory, and peripheral devices. The kernel is the "lowest" level of the OS. It is responsible for all major activities of this operating system. Linux kernel is written in the version of the C programming language.
3. **Shell:** Shell acts as a user interface, interpreting user commands and starting an application. Users typically interact with a Linux shell using a **terminal emulator**, however, direct operation via serial hardware connections, or networking session, are common for server systems.
4. **System utilities/Application:** The system tools are built using the system libraries and enable administrators to administer the system manage processes, navigate on the file system, execute other applications, configure the network etc..

# Linux Components

- **Boot-loader-** It is a program that loads the Linux kernel into the computer's main memory, by being executed by the computer when it is turned on and after the firmware initialization is performed.
- **Kernel-** It is the core part of the operating system and manages the CPU, memory, and peripheral devices.
- **Daemons-** These are background services that either startup during boot, or after you log into the desktop.
- **Shell-** Shell is a command processor that allows you to control the computer via commands typed into a text interface.
- **System Libraries-** System libraries are special programs that help in accessing the kernel's features. Programmers have developed a standard library of procedures to communicate with the kernel.
- **Graphical Server-** This is the sub-system that displays the graphics on your monitor. It is commonly referred to as the X server or just "X".
- **System Tools-** Linux OS has a set of utility tools which are usually simple commands. It is a software which GNU project has written and publish under their open source license so that software is freely available to everyone.

# Linux Distributions

- A Linux distribution (distro) is an operating system made from a software collection, which is based upon the Linux kernel and a package management system.
- A typical Linux distribution is a collection of Linux kernel, system libraries, GNU tools, additional applications, documentation, a window system manager and a desktop environment.
- There are almost six hundred Linux distributions exist and with close to five hundred out of those in active development.
- There are commercially backed distributions, such as **Fedora** (Red Hat), **openSUSE** (SUSE) and **Ubuntu** (Canonical Ltd.), and entirely community-driven distributions, such as **Debian**, **Slackware**, **Gentoo**, and **Arch Linux**.

# Linux Kernel

- The Linux Kernel is the heart of the operating system.
- It is mainly responsible for the software and hardware of our computer working correctly and can interact with each other.
- Linux has **monolithic kernel,** there is no different memory used for user services and kernel services. the same memory space is used to implement user services and kernel services.
- As it uses the same memory space, the size of the kernel increases, increasing the overall size of the OS.
- The main functions that the Kernel performs are as follows:
  1. Process Management-The creation, execution, and termination of processes keep on going inside the system whenever a system is in the ON mode.
  2. Memory Management- The kernel has full control for accessing the computer's memory.
  3. Device Management- The Kernel also manages all the different devices which are connected to the system, like the Input and Output devices, etc.
  4. Interrupt Handling- While executing the processes, there are conditions where tasks with more priority need to be handled first.
  5. Input Output Communication- it is also responsible for handling all sorts of input and output that is exchanged through these devices.

# Revision Questions

1. Differentiate between Linux and Windows Operating Systems
2. Explain any FIVE Linux features that makes it a very popular operating system nowadays.
3. Define Linux kernel and explain FOUR functions performed by the kernel.
4. With aid of a diagram, discuss the Linux Architecture.
5. Give FOUR examples of Linux distributions.